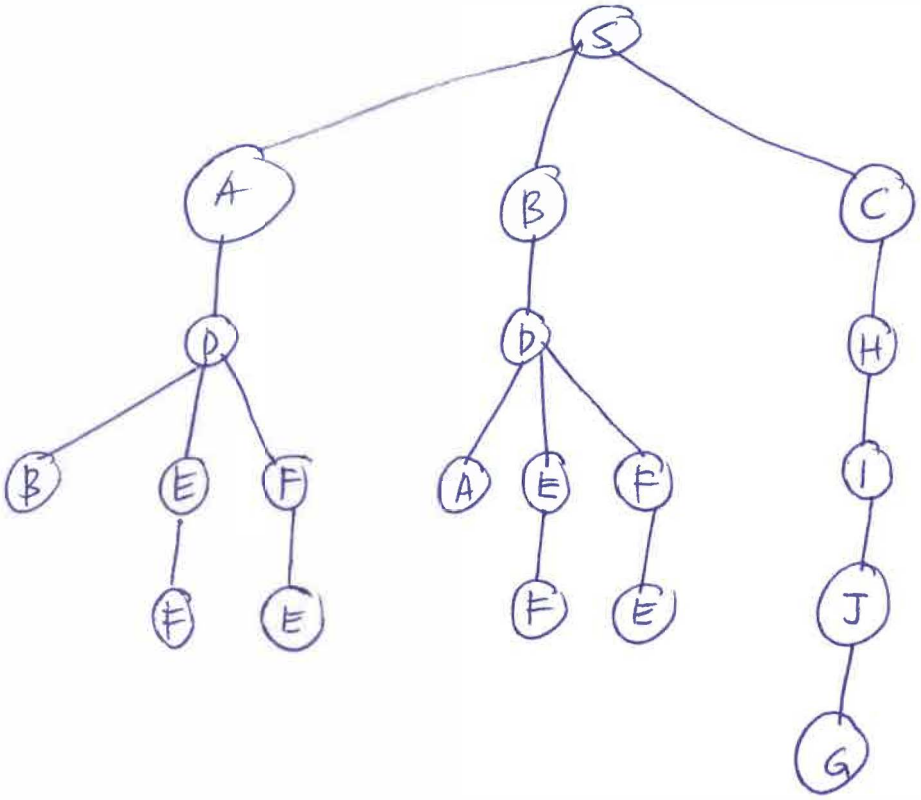
Search Tree



Heuristics

S = 10

A = 5

B = 8

C = 6

D = 7

E = 6

F = 5

H = 3

I = 2

J = 1

G = 0

1) DFS with Visited

Step	Pop	Enq	Visited list
0		(S)	S
1	(S)	~~(AS)~~ (BS) (CS)	S, A, B, C
2	(AS)	~~(DAS)~~ (BS) (CS)	", D
3	(DAS)	~~(EDAS)~~ (FDAS) (BS) (CS)	", E, F
4	(EDAS)	~~(FDAS)~~ (BS) (CS)	"
5	(FDAS)	~~(BS)~~ (CS)	"
6	(BS)	~~(CS)~~	"
7	(CS)	~~(HCS)~~	", H
8	(HCS)	~~(IHCS)~~	", I
9	(IHCS)	~~(JIHCS)~~	", J
10	(JIHCS)	~~(GJIHCS)~~	", G
11	(GJIHCS)		

Seq. of State Exp: S-A-D-E-F-B-C-H-I-J-G

path: S-C-H-I-J-G

max Q = 4

state expansions = 11

2) DFS without Visited List

Step	Dequened	Enqueued
0		(S)
1	(S)	(AS)(BS)(CS)
2	(AS)	(DAS)(BS)(CS)
3	(DAS)	(BDAS)(EDAS)(FDAS)(BS)(CS)
4	(BDAS)	(EDAS)(FDAS)(BS)(CS)
5	(EDAS)	(FEDAS)(FDAS)(BS)(CS)
6	(FEDAS)	(FDAS)(BS)(CS)
7	(FDAS)	(EFDAS)(BS)(CS)
8	(EFDAS)	(BS)(CS)
9	(BS)	(DBS)(CS)
10	(DBS)	(ADBS)(EDBS)(FDBS)(CS)
11	(ADBS)	(EDBS)(FDBS)(CS)
12	(EDBS)	(FEDBS)(FDBS)(CS)
13	(FEDBS)	(FDBS)(CS)
14	(FDBS)	(EFDBS)(CS)
15	(EFDBS)	(CS)
16	(CS)	(HCS)
17	(HCS)	(IHCS)
18	(IHCS)	(JIHCS)
19	(JIHCS)	(GJIHCS)
20	(GJIHCS)	

Fin

Sequence of State expansion : S-A-D-B-E-F-F-E-B-D-A-E-F-F-E-C-H-I-J-G

3) BFS w/ Visited List

Step	Deq	Enq	Visited List
0		~~(S)~~	S
1	(CS)	~~(AS)~~(BS)(CS)	S, A, B, C
2	(AS)	~~(BS)~~(CS)(DAS)	", D
3	(BS)	~~(CS)~~(DAS)	"
4	(CS)	~~(DAS)~~(HCS)	", H
5	(DAS)	~~(HCS)~~(EDAS)(FDAS)	", E, F
6	(HCS)	~~(EDAS)~~(FDAS)(IHCS)	", I
7	(EDAS)	~~(FDAS)~~(IHCS)	"
8	(FDAS)	~~(IHCS)~~	"
9	(IHCS)	~~(JIHCS)~~	", J
10	(JIHCS)	~~(GJIHCS)~~	", G
11	(GJIHCS)		

Sequence of State Exp: S - A - B - C - D - H - E - F - I - J - G

Num of State Expansions = 11

Max Q length = 3

path: S - C - H - I - J - G

path length: 5

4) BFS without Visited list

Step	Deq	Enq
0		(S)
1	(S)	~~(AS)~~ (BS) (CS)
2	(AS)	~~(BS)~~ (CS) (DAS)
3	(BS)	~~(CS)~~ (DAS) (DBS)
4	(CS)	~~(DAS)~~ (DBS) (HCS)
5	(DAS)	~~(DBS)~~ (HCS) (BDAS) (EDAS) (FDAS)
6	(DBS)	~~(HCS)~~ (BDAS) (EDAS) (FDAS) (ADBS) (EDBS) (FDBS)
7	(HCS)	~~(BDAS)~~ (EDAS) (FDAS) (ADBS) (EDBS) (FDBS) (IHCS)
8	(BDAS)	~~(EDAS)~~ (FDAS) (ADBS) (EDBS) (FDBS) (IHCS)
9	(EDAS)	~~(FDAS)~~ (ADBS) (EDBS) (FDBS) (IHCS) (FEDAS)
10	(FDAS)	~~(ADBS)~~ (EDBS) (FDBS) (IHCS) (FEDAS) (EFDAS)
11	(ADBS)	~~(EDBS)~~ (FDBS) (IHCS) (FEDAS) (EFDAS)
12	(EDBS)	~~(FDBS)~~ (IHCS) (FEDAS) (EFDAS) (FEDBS)
13	(FDBS)	~~(IHCS)~~ (FEDAS) (EFDAS) (FEDBS) (EFDBS)
14	(IHCS)	~~(FEDAS)~~ (EFDAS) (FEDBS) (EFDBS) (JIHCS)
15	(FEDAS)	~~(EFDAS)~~ (FEDBS) (EFDBS) (JIHCS)
16	(EFDAS)	~~(FEDBS)~~ (EFDBS) (JIHCS)
17	(FEDBS)	~~(EFDBS)~~ (JIHCS)
18	(EFDBS)	~~(JIHCS)~~
19	(JIHCS)	~~(GJIHCS)~~
20	(GJIHCS)	

5) Best-First Search with Visited List

Step	Deq	Enq	Visited List
0		~~(10S)~~	S
1	(10S)	~~(5AS)~~ (8BS) (6CS)	S, A, B, C
2	(5AS)	(7DAS) (8BS) ~~(6CS)~~	", D
3	(6CS)	~~(3HCS)~~ (7DAS) (8BS)	", H
4	(3HCS)	~~(2IHCS)~~ (7DAS) (8BS)	", I
5	(2IHCS)	~~(1JIHCS)~~ (7DAS) (8BS)	", J
6	(1JIHCS)	~~(0GJIHCS)~~ (7DAS) (8BS)	", G
7	(0GJIHCS)		